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Code Understanding Journal

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Part 1: Task Creation & Status Updates

Main Components Involved

- **cli.py (Command Line Interface):** Reads user inputs, parses flags (--description, --priority, --due, --tags), and routes requests to the manager layer.
- **task_manager.py (Service Layer):** Handles core business logic, assigns unique IDs, and validates inputs.
- **models.py (Domain Models):** Defines the Task object along with Enums like TaskStatus (TODO, IN_PROGRESS, REVIEW, DONE) and TaskPriority.
- **repository.py (Storage Layer):** Reads tasks from a local JSON file into memory and writes modified tasks back to disk.

Execution Flow

Task Creation Flow:

1. User enters command in CLI.
2. cli.py parses arguments (title, description, priority, due date, tags).
3. task_manager.create_task() creates a new Task instance with a unique ID.
4. repository.save_task() converts the Task object to JSON format.
5. Task is saved directly to tasks.json.

Status Update Flow:

1. User runs python cli.py update-status <task_id> <new_status>.
2. cli.py parses the ID and validates the status string.
3. task_manager.update_status() finds the task by ID and changes task.status.
4. repository.save_all() overwrites tasks.json with the updated task list.

Storage and Retrieval

- **Storage:** Kept in a lightweight tasks.json file on disk.
- **Retrieval:** On startup, the repository reads tasks.json, parses the raw JSON text using Python's built-in json module, and converts dictionary data back into live Task objects.

Key Design Patterns

- **Repository Pattern:** Separates file storage logic from core business logic.
- **Command Routing Pattern:** cli.py acts as a controller, delegating work to

task_manager.py instead of executing business rules directly.

Part 2: Task Prioritization System

Initial Understanding vs. Discoveries

- **Priority Values:**
 - *Initial Guess:* Stored as plain numbers (1, 2, 3, 4).
 - *Discovered:* Encapsulated in a TaskPriority Enum (1=LOW, 2=MEDIUM, 3=HIGH, 4=URGENT).
- **Filtering:**
 - *Initial Guess:* cli.py manually iterates through list items.
 - *Discovered:* TaskManager uses a dedicated get_tasks_by_priority() service method.
- **Validation:**
 - *Initial Guess:* Allowed any integer input.
 - *Discovered:* Invalid levels are rejected by Enum validation before reaching storage.

Key Insights & Clarified Misconceptions

- Converting raw CLI integers to Enums prevents invalid data from corrupting storage.
- Filtering by numerical Enums simplifies logic (e.g., checking if priority >= 3).
- **Clarified Misconception:** Priority is not just visual label text printed in the terminal—it drives real business behavior (such as protecting urgent tasks from cleanup routines).

Part 3: Data Flow & State Management

Complete Data Flow (Marking Task as Complete)

1. **User Action:** Command executed: python cli.py update-status 5 done
2. **CLI Layer:** cli.py converts "5" to integer 5 and maps "done" to TaskStatus.DONE.
3. **Manager Layer:** TaskManager.update_status(5, TaskStatus.DONE) locates task ID 5 and sets task.status = TaskStatus.DONE.
4. **Persistence Layer:** TaskRepository.save() converts the updated Task object into a dictionary and writes the payload to tasks.json.
5. **Output:** CLI displays a success message to the terminal.

State Changes & Failure Points

- **State Change:** Task.status updates from TODO / IN_PROGRESS / REVIEW directly to TaskStatus.DONE.
- **Potential Failure Points:**
 - *Non-Existent Task ID:* Raises TaskNotFoundError.
 - *Invalid Status String:* Fails Enum validation (e.g., typing "finished" instead of "done").
 - *Storage Issue:* Read-only or locked tasks.json file triggers an IOError.

Part 4: Presentation Outline

- **Slide 1: Architecture Overview** — Clean separation of concerns (CLI, Business Logic,

Models, Storage) with zero third-party dependencies.

- **Slide 2: Core Feature Flow** — How task creation, priority filtering, and status updates move safely through each application layer.
- **Slide 3: Design Pattern Spotlight** — The Repository Pattern keeps storage logic independent and easy to test or replace.
- **Slide 4: Key Reflections** — Using structured AI prompts helps quickly map data flow and clarify domain models without relying on complex IDE debuggers.